

■ 1.9.2 C and FORTRAN Programming Languages

Mathematica is a system for doing general mathematical computations. You may sometimes want to take results you have got from *Mathematica*, and plug them into various special-purpose programs you may have written in other programming languages.

The standard version of *Mathematica* supports both C and FORTRAN programming languages. You can have *Mathematica* write out expressions in the form you need for input to C and FORTRAN.

<code>CForm[expr]</code>	write out <i>expr</i> so it can be used in a C program
<code>FortranForm[expr]</code>	write out <i>expr</i> for FORTRAN

Mathematica output for programming languages.

Here is an expression, written out in standard mathematical form.

```
In[1]:= Expand[(1 + x + y)^2]
Out[1]= 1 + 2 x + x^2 + 2 y + 2 x y + y^2
```

Here is the expression in FORTRAN form.

```
In[2]:= FortranForm[%]
Out[2]//FortranForm= 1 + 2*x + x**2 + 2*y + 2*x*y + y**2
```

Here is the same expression in C form. Macros for objects like `Power` are defined in the C header file `mdefs.h` that comes with most versions of *Mathematica*.

```
In[3]:= CForm[%]
Out[3]//CForm=
1 + 2*x + Power(x,2) + 2*y + 2*x*y + Power(y,2)
```